CUNEF UNIVERSIDAD

Quantitative Methods for Humanities

Causality

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Introduction to Causality
a Historical Approach

Why study causality?

- The notion of **cause and effect** is central in many disciplines:
  - **Economics:** Does an increase in the minimum wage cause unemployment to rise?
  - **Politics:** Would universal health care improve the health and finances of the poor?
  - **Philosophy:** Does belief in God cause moral behavior?
  - **Health:** What is the impact of the use of social media in the feeling of loneliness?
  - **Environmental:** Is deforestation worsening global warming?

> We do not have knowledge of a thing until we have grasped its why, that is to say, its cause.
>
> - Aristotle, Physics

But what do we mean when we say *cause and effect*? Or, in other words

What is Causality?

What is causality?

David Hume's[1] **conditions of causality**[2]

Regularity: A cause is something that is regularly followed by an effect
Temporality: The cause must occur before the effect in time.
Necessity: If there is no cause there is no effect

> We may define a cause to be an object followed by another, and where all the objects, similar to the first, are followed by objects similar to the second, [...] if the first object had not been, the second never had existed
>
> - David Hume, An Enquiry Concerning Human Understanding

[1] Scottish philosopher, historian, and economist regarded as the father or modern causality theory.
[2] David Hume (1757) An Enquiry Concerning Human Understanding.

What is (regularity) causality

Hume's **Regularity Theory of Causality**[1,2]

Principle: Focuses on Hume's regularity condition.
Method: Empirical observation.
Regular Causality: An event C (cause) always precedes an event E (effect).

Example: Sunrise and daylight

Regularity: Every time the sun rises (C) daylight (E) follows.
Causality: Therefore sunrise causes daylight.

[1] David Hume (1739) A Treatise of Human Nature
[2] John Stuart Mill (1843) A System of Logic, Ratiocinative and Inductive: Being a Connected View of the Principles of Evidence, and the Methods of Scientific Investigation

What is (counterfactual) causality?

David Lewis's **Counterfactual Causality**[1]

Principle: Focuses on Hume's necessity condition.
Actual World: Events C (presumed cause) and E (presumed effect) have occurred.
Possible No-C World: Hypothetical version of the actual world, where C has not occur
Closest No-C Possible World: The change required to eliminate C is minimal.
Counterfactual Causality: If E does not occur in the closest possible world.

Example: A rock breaks a window

Actual World: The rock is thrown (C) and the window shatters (E)
Possible No-C World: The person holding the rock decides not to throw it.
Closest No-C Possible World: No other object impacts the window.
Counterfactual Causality: The window does not break.

[1] David K. Lewis (1973) Counterfactuals. The Journal of Symbolic Logic.

Issues with deterministic causality

The regularity and counterfactual approaches, as originally posed, **sometimes fail**

Example: Common cause

Regularity approach: Increased ice cream sales cause more drownings because they are regularly observed together in summer.

Example: Probabilistic nature

Regularity approach: A drug killing a patient is not a causal effect as it is not regularly observed.

Example: Overdetermination

Premises:

- A prisoner is executed (E) by two soldiers.
- Soldiers (C_1) and (C_2) fire simultaneously, and either shot is sufficient to kill.

Counterfactual approach:

- **Analysis:** If C_1 hadn't fired, C_2 would still have killed the prisoner, and vice versa.
- **Result:** Neither C_1 nor C_2 cause E.

What is (probabilistic) causality?

Hans Reichenbach's **Probabilistic Causality**[1]

Premises:

- Causes not always produce effects
  (e.g., smoking does not always cause lung cancer).
- Effects may have multiple causes
  (e.g., lung cancer can result from smoking, pollution, or genetic predisposition).

Probabilistic Causation: Event C causes event E if it increases its probability

- i.e., $\mathbb{P}(E \mid C) \geq \mathbb{P}(E)$
- e.g., Smoking raises lung cancer risk

Screening off effect: A common cause explains the correlation between two events e.g., Smoking causes both yellow fingers and lung cancer, but yellow fingers does not cause lung cancer or vice versa and.

[1] Hans Reichenbach (1956) The Direction of Time. The Philosophical Quarterly.

When causality goes wrong

Getting causality wrong might have severe consequences

- Cannabis and "gateway drug effect"[1]
- Vaccines and autism[2]
- Death penalty and homicide rates[3]

> A fool can throw a stone in a pond that 100 wise men cannot get out.
>
> - Saul Bellow

[1] William. Noel and Judy Wang (2018) Is Cannabis a Gateway Drug? Key Findings and Literature Review.
[2] Michael Davidson (2017) Vaccination as a cause of autism—myths and controversies.
[3] National Research Council (2012) Deterrence and the Death Penalty

Understanding Causality. A modern approach

An illustrative case study

To illustrate the goal of **causal inference** and lay the ground to introduce the **potential outcome framework**, we ask ourselves the following questions:

- Does racial discrimination exist in the labor market?
- Is this a discrimination due to race perse, or should it be attributed to other factors such as racial gaps in educational attainment?

To answer, imagine that we have a dataset of $n \in \mathbb{N}$ résumés sent to employers with the following variables

- perceived race of the job applicant (white-sounding or black-sounding name)
- whether a callback was made by the employer (yes or no).
- education level, gender, and other *pretreatment variables*.

Treatment and outcome.

- **Treatment Variable (T):**[1] Independent (typically binary) variable that is deliberately manipulated to observe its effect on the outcome. (e.g., résumé applicant's name)
  - $T = 1$ if the treatment is applied (e.g., black-sounding name)
  - $T = 0$ if the is not applied (e.g., white-sounding name)
- **Outcome Variable (Y):**[2] Dependent variable that is measured to determine the effect of the treatment. (e.g., $Y = 1$ if the employer calls back, $Y = 0$ otherwise)

[1] **Alternative names:** exposure, intervention, independent, cause variable.
[2] **Alternative names:** response, dependent, effect variable.

Sample. Control and treatment groups

- **Individual metrics:** There are $n \in \mathbb{N}$ total individuals in the study, each associated to the treatment T_i and outcome Y_i, for $i = 1, \ldots, n$.
- **Treatment group:** Set of individuals who **received the treatment**, e.g., set of employers who were sent black-sounding names.
- **Control group:** Set of individuals who **do not received the treatment**, e.g., set of employers who were sent white-sounding names.

The stable unit treatment value assumption

The stable unit treatment value assumption (SUTVA)

The **Stable Unit Treatment Value Assumption (SUTVA)** is a key assumption in causal inference, particularly in the Potential Outcome Framework.

SUTVA

The SUTVA states:

- **No Interference/Spillovers Assumption (NI):** the potential outcomes for any unit do not vary with the treatments assigned to other units.
- **No Hidden Variants (NHV):** for each unit, there are no different forms or versions of each treatment level, which lead to different potential outcomes.

The SUTVA: no interference/spillovers

- **Mathematical interpretation:** mathematically, it meas that

$$\mathbb{P}\left(Y_i \leq y \mid T_1, \ldots, T_n\right) = \mathbb{P}\left(Y_i \leq y \mid T_i\right).$$

Example:

- **Scenario:** Comparing callback rates for résumés with black-sounding names (treatment) vs. white-sounding names (control).
- **Compliance case:** an individual's callback decision does not depend on the name on other résumés.
- **Non-compliance case:** an employer who receives several résumés with white-sounding names may become more biased and less likely to call presumably back applicants.

What would happen without the NI assumption. Accounting for interaction

- **Without NI:** You may need to account for cross interaction
- **Example:** Suppose that
  - The résumés are sent in pairs to the same employer.
  - You want to estimate the callback average difference between sending black-black vs white-black sounding names.

$$\mathbb{E}\left[Y_1 | T_1 = 1, T_2 = 1\right] - \mathbb{E}\left[Y_1 | T_1 = 1, T_2 = 0\right].$$

- **Drawbacks:**
  - The conclusion is less general.
  - Requires higher dimensional data.

What would happen without the NI assumption. Layering

- **Without NI:** You may need to consider groups where NI hold
- **Example:** You want to test if reduced class sizes causes improved student performance in mathematics.
  - **Problem:** It is not sensible to assume that there is no interference among students within the same school or classroom.
  - **Solution:** Assume NI between different school/classrooms.
- **Drawbacks:**
  - The conclusion is less general.
  - Requires higher data sizes.

The SUTVA: no hidden variants

- **Interpretation:** Under the NHV assumption, the possible implementations of the treatment are irrelevant for the outcome of one individual. That is, **the outcome depends only on the treatment level** (0 or 1 for binary treatments) and not on how this treatment was implemented.
- **Mathematical interpretation:** mathematically, this assumption means that the outcome is a (random) function of the treatment

$$Y_i = Y_i(T_i).$$

For binary treatments:

$$Y_i = \begin{cases} Y_i(0), & \text{if } T_i = 0 \\ Y_i(1), & \text{if } T_i = 1 \end{cases}.$$

The SUTVA: no hidden variants. An example

- **Scenario:** Comparing callback rates for résumés with black-sounding names (treatment) vs. white-sounding names (control).
- **Compliance case:** All résumés in the treatment group and control group are identical, only varying the name.
- **Non-compliance case:** Some résumés with black-sounding names are different in format (e.g., adding extra details, changing the style), or vary significantly in education levels, which might lead to different outcomes.

What would happen without the NHV assumption?

- The NHV assumption allows to identify the observed outcome as the outcome under or without the treatment.
- Without the NHV assumption we just observe outcome attached to the specific version of the treatment that was applied.

With the NHV assumption

T_i	$Y_i(1)$	$Y_i(0)$
1	Y_i	?
0	?	Y_i

Without the NHV assumption

T_i	Y_i
1	Y_i
0	Y_i

NHV allows to introduce factuals and counterfactuals in a simple and powerful way!

Potential outcomes, factual and counterfactual. Under NHV

Under the NHV assumption:

- **Potential Outcomes:**
  - $Y(1)$**:** Outcome under treatment. (e.g, callback if black-named applicant)
  - $Y(0)$**:** Outcome without treatment. (e.g., callback if white-named applicant)
- **Factual outcome:** Observable outcome (what happened).

$$Y(T)$$

- **Counterfactual outcome:** Unobservable outcome (what would have happened if the treatment were different).

$$Y(1-T) \quad \text{(for binary treatments).}$$

Potential outcomes, factual and counterfactual. Without NHV

Without the NHV assumption:

- **Potential Outcomes:**
  - $Y(1, j)$**:** Outcome under variation j of the treatment, for $j = 1, \ldots, v$.
  - $Y(0, j)$**:** Outcome without variation j of the treatment, for $j = 1, \ldots, v$.
- **Factual outcome:** Observable outcome (what happened).

$$Y(T, j)$$

- **Counterfactual outcome:** Unobservable outcome (what would have happened).

$$Y(1 - T, j) \quad \text{(for binary treatments)}.$$

Conclusion: You will need increasingly higher sample size as the number of variants increases. Impracticable for infinite number of variants.

The Potential Outcome Framework

The Potential Outcome Framework[1]

The **Potential Outcome Framework** (POF), also known as the Rubin Causal Model, is a widely used approach in **causal inference** that helps to understand and estimate **causal effects**, based on the idea of comparing **potential outcomes**.

Rubin's definition of causal effect: "Intuitively, the causal effect of one treatment, E, over another, C, is the difference between **what would have happened if** the unit had been exposed to E and what would have happened if the unit had been exposed to C"

Rubin's example of potential outcomes: "If an hour ago I had taken two aspirins instead of just a glass of water, my headache would now be gone, or because an hour ago I took two aspirins instead of just a glass of water, my headache is now gone."

[1] Donald B Rubin (2006) Causal Inference Using Potential Outcomes: Design, Modeling, Decisions.

Individual causal effect and the fundamental problem of causal inference

- **Individual treatment (causal) effect:** difference between the two potential outcomes:

$$\mathrm{TE} = Y(1) - Y(0).$$

Fundamental Problem of Causal Inference[1] (FPCI)

We observe only one of the two potential outcomes (the factual outcome), so we never get to observe the effect of the treatment on the outcome.

- **Individual causal inference:** The FPCI states that it is impossible to estimate the individual causal effect. You only have one non-observable observation for this phenomenon!

[1] Paul W. Holland (1985) Statistics and Causal Inference.

An example

Table: **Potential Outcome Framework and Fundamental Problem of Causal Inference.**

Résumé	*Black-sounding*	*Callback*		*Age*	*Education*
i	name T_i	$Y_i(1)$	$Y_i(0)$		
1	1	1	?	20	college
2	0	?	0	55	high school
3	0	?	1	40	graduate school
⋮	⋮	⋮	⋮	⋮	⋮
n	1	0	?	62	college

Average causal effect

- **Average causal inference:** Despite the FPCI, it is possible to estimate the causal effect on average terms.
- **Average treatment (causal) effect:** Aggregated measure of the treatment effect across all the individuals in a population. Defined as the expectation of the individual treatment effect:

$$\mathrm{ATE} = \mathbb{E}\left[\mathrm{TE}\right] = \mathbb{E}\left[Y(1) - Y(0)\right].$$

The Sample Average Treatment Effect (SATE)

The **sample average treatment effect** is the sample average of individual-level causal effects:

$$\mathrm{SATE} = \overline{\mathrm{TE}} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{TE}_i = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i(1) - Y_i(0)\right) = \overline{Y(1)} - \overline{Y(0)},$$

where n is the sample size and the subindex i represents the ith observation.

Non observability of SATE

Problem: The sample average response cannot be directly observed.

$$\overline{Y(0)} = \frac{1}{n}\sum_{i=1}^{n} \underbrace{Y_i(T_i)\mathbb{1}(T_i = 0)}_{\text{observable (factual)}} + \frac{1}{n}\sum_{i=1}^{n} \underbrace{Y_i(1 - T_i)\mathbb{1}(T_i = 1)}_{\text{non-observable (counterfactual)}},$$

$$\overline{Y(1)} = \frac{1}{n}\sum_{i=1}^{n} \underbrace{Y_i(1 - T_i)\mathbb{1}(T_i = 0)}_{\text{non-observable (counterfactual)}} + \frac{1}{n}\sum_{i=1}^{n} \underbrace{Y_i(T_i)\mathbb{1}(T_i = 1)}_{\text{observable (factual)}}.$$

We need to **estimate the missing parts**!

Estimating the SATE

Depending on the **data type** and its **underlying circumstances**, many methods can be used to **estimate the SATE**. Among the most common ones are:

- Difference-in-Means (DiM) estiamator
- Before-and-after (BA) estimator
- Difference-in-differences (DiD) estimator

Choosing the right estimator (next chapter) is critical to **avoid bias in the estimation**.

Sources of bias

Pretreatment variables and Confounders

Pretreatment variable: measured before the treatment or intervention is applied (e.g., age, education, gender,...)

Confounders[1]

A pretreatment variable that is associated with both the treatment and the outcome variables is called a confounder and is a **source of confounding bias** in the estimation of the treatment effect.

[1] **Alternative names:** confounding factor, common causes.

Confounding bias

Confounding bias

Distortion of the estimated relationship between the treatment and the outcome due to the influence of a confounder

Example:

- **Correlation with the treatment:** the résumés of white-sounding names may be associated with higher education status.
- **Correlation with the outcome:** higher education status may positively influence callback rates.
- **Confounder:** Education status may be a confounder.

Selection bias

Selection bias

Phenomenon that occurs when the sample of individuals selected for a study is not representative of the broader population, leading to biased estimates.

Example:

- **Non-representative sample:** Sending only résumés with high levels of education.
- **Selection bias:** the results may not reflect the discrimination faced by less-educated individuals.

Types of Selection Bias

- **Sampling Bias:** When certain individuals are more likely to be selected, leading to a non-representative sample.
- **Self-Selection Bias:** When participants choose themselves into a study, potentially skewing results.
- **Attrition Bias:** When participants drop out of a study, affecting the representativeness of the remaining sample.
- **Non-Response Bias:** When those who don't respond differ systematically from those who do.
- **Time Interval Bias:** When data collection timing affects the results.
- **Survivorship Bias:** When only successful subjects are considered, overlooking failures.
- **Common Source Bias:** When both variables are collected from the same group, causing shared biases.

Avoiding bias through study designs

Question: In the study about the racial discrimination in the job market, would you say that there is confounding and/or selection bias?

Answer: we need more information about how the study was conducted.

In general, **how a research is conducted could dramatically affect the way we interpret the results.**

We will dive deeper into this topic in the
Next chapter!